

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Interdisciplinary Project Work

on

“SMART ATTENDANCE SYSTEM”

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Interdisciplinary Project-Based Learning under the **Innovation & Design Thinking**
Laboratory,
Department of Basic Sciences.

by

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DEPARTMENT OF BASIC SCIENCES



CERTIFICATE

This is to certify that the File Structures mini project entitled “SMART ATTENDANCE SYSTEM” has been successfully carried out by **Ganesh P(1CR25AD041), G Vinayak(1CR25AD042), Sumith S(1CR25IS174), Surabhi M(1CR25IS175)** bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Interdisciplinary Project-Based Learning, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

Signature of Guide

**Prof.Manasa
Asst. Professor**

Signature of HOD

**Dr. Fazlur Rahaman
HoD,
Dept. of Chemistry,
CMRIT**

External Viva

Name of the examiners

Signature with date

1.

2.

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Abstract

The Smart Attendance System is a software-based project developed to automate and simplify the process of attendance management in educational institutions. The main objective of the project is to reduce manual effort, prevent proxy attendance, and improve the accuracy of attendance records through digital automation. The system uses face recognition technology to identify students and automatically mark attendance using a webcam and a centralized database. Various technologies such as Python, OpenCV, database management systems, and web development tools were used for designing and implementing the system. The project includes modules for student registration, face data collection, attendance marking, and report generation. Testing and user reviews were conducted to evaluate the performance and usability of the system. The developed prototype successfully demonstrated efficient attendance tracking, reduced time consumption, and improved data management compared to traditional attendance methods. The project also highlighted the practical application of artificial intelligence and automation in educational environments while providing a user-friendly and secure attendance solution.

Chapter 1

INTRODUCTION

Attendance management is an important process in educational institutions and organizations. Traditional attendance methods such as paper registers are time-consuming, prone to errors, and allow proxy attendance. To overcome these problems, the Smart Attendance System was developed using face recognition technology. The system automatically detects and recognizes faces through a camera and marks attendance digitally in a database. Technologies such as Python, OpenCV, and database management systems are used for implementation. The project improves accuracy, reduces manual effort, saves classroom time, and simplifies attendance record management. It also demonstrates the practical application of automation and biometric technology in modern education systems.

1.1 Brief history of attendance system

Attendance systems have been used for many years in schools, colleges, and workplaces to record the presence of students and employees. Earlier, attendance was maintained manually using paper registers, where teachers or supervisors marked attendance by calling names or checking signatures. Although this method was simple, it consumed a lot of time and was prone to errors, manipulation, and proxy attendance. With technological advancements, institutions gradually shifted to electronic attendance systems such as punch cards, barcode scanners, and RFID-based systems. These systems reduced manual work and improved data storage but still faced issues like card sharing and lack of security. The increasing need for automation, accuracy, and security led to the development of smart attendance systems that use biometric technologies and artificial intelligence for better attendance management.

1.2 Modern Biometric system

The use of face recognition for attendance has become one of the common applications of modern biometrics due to the contact-free nature of the method, as well as its speed and accuracy. Unlike traditional approaches like manual registers, ID card readers, and fingerprint scanners that identify people based on fingerprints, facial biometric technology identifies an individual on the basis of unique features of the face and does not require any physical interaction with the device. This increases convenience and hygiene while making it convenient for use by many people at a time, for example in educational institutions or companies.

To perform identification using face recognition, a video camera detects the face of a particular person and compares it to the pre-saved facial data available in the database. With the help of various algorithms and machine learning tools, the software can detect faces from images and videos in real time and automatically register attendance along with other parameters like date and time.

Using biometrics for attendance has many advantages compared to traditional ways of tracking attendance.

Moreover, developments in the fields of Artificial Intelligence and computer vision have enhanced the effectiveness and dependability of facial recognition software even with varied lighting effects, facial expressions, and camera angles. It is due to the advantages offered by them that biometric-based systems are becoming increasingly popular in academic institutions, corporate offices, and other secure premises for automation of attendance management systems.

Chapter 2

Problem Statement

2.1 Description

Attendance management in educational institutions is still commonly performed using manual methods such as paper registers and signature sheets. These traditional systems consume valuable classroom time, require extensive manual effort, and are prone to errors and data mismanagement. In many cases, proxy attendance and inaccurate record maintenance reduce the reliability of the attendance process. Managing and calculating attendance records manually also becomes difficult when handling a large number of students. Therefore, there is a need for an automated, secure, and efficient attendance management system that can improve accuracy, reduce manual work, and simplify record maintenance.

2.2 Challenge Statement

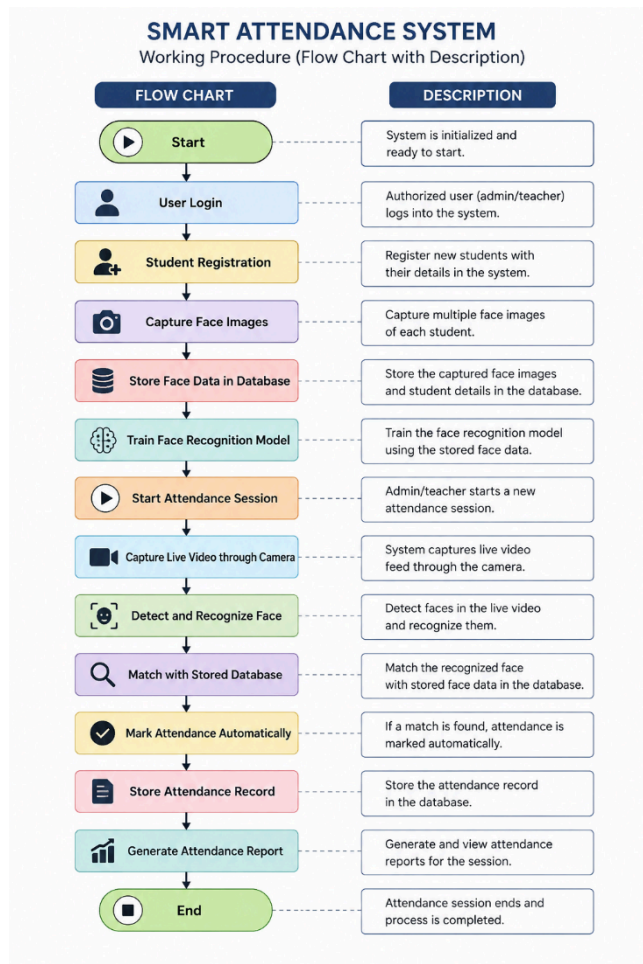
The major challenge is to develop a smart attendance system that can automatically identify students and mark attendance accurately using biometric face recognition technology. The system must reduce proxy attendance, ensure secure data storage, provide real-time attendance updates, and maintain high recognition accuracy under different lighting and environmental conditions. Additionally, the system should be user-friendly, reliable, and capable of handling attendance data efficiently for educational institutions.

Chapter 3

3.1 Design Thinking Process

- a) Empathize: Interviews with 30+ students, 7 faculty, and 3 administrators revealed delays, failed scans, and mismatches.
- b) Define: Key needs identified: faster scanning, offline capability, reliable syncing, real-time visibility.
- c) Ideate: Generated 10+ ideas. Final solution selected: biometric + IoT device with dashboard.
- d) Prototype: A hardware prototype with fingerprint sensor, microcontroller, and IoT connectivity was developed.
- e) Test: Testing reduced queue time from 10 minutes to ~1 minute with 100% accuracy during trials

3.2 Methodology



3.3 Prototype Description

3.3.1 Materials Used (With specific images, descriptions and specifications

3.3.1 Materials Used

(With Descriptions and Specifications)

1. Laptop/Desktop System



- **Used for developing and running the Smart Attendance System.**
- **Performs face recognition processing and database management.**

Specifications:

- **Processor: Intel i5 / Ryzen 7**
- **RAM: Minimum 8 GB**
- **Storage: 256 GB SSD**
- **Operating System: Windows**

2. Webcam / Camera Module



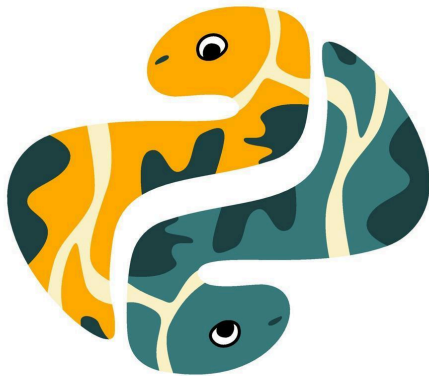
- **Captures live facial images of students.**

- **Used for face detection and recognition.**

Specifications:

- **Resolution: 720p / 1080p**
- **Frame Rate: 30 FPS**

3. Python Programming Language

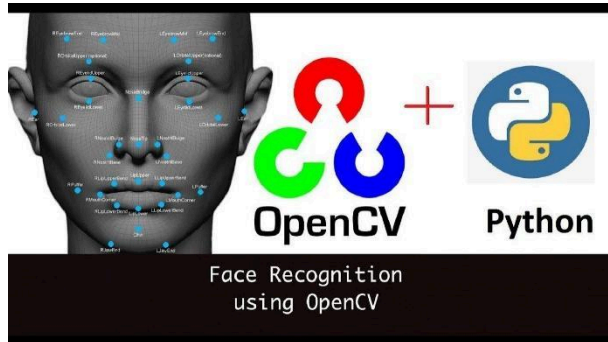


- **Main programming language used for system development.**
- **Handles backend processing and automation.**

Specifications:

- **Version: Python 3.x**
- **Libraries: OpenCV, NumPy, Pandas**

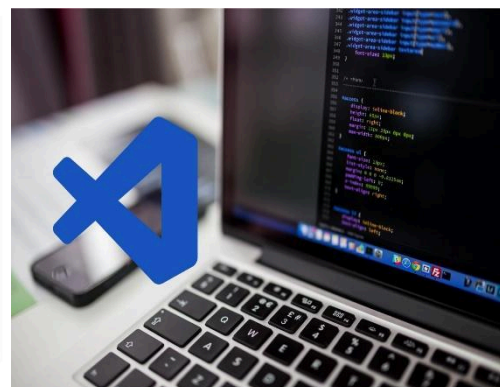
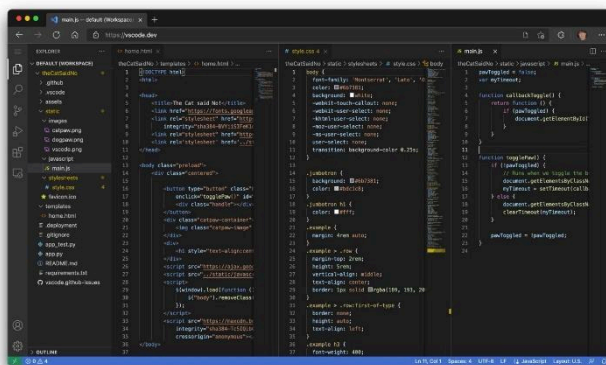
4. OpenCV Library



- Used for image processing and face recognition.
- Detects and identifies faces in real time.

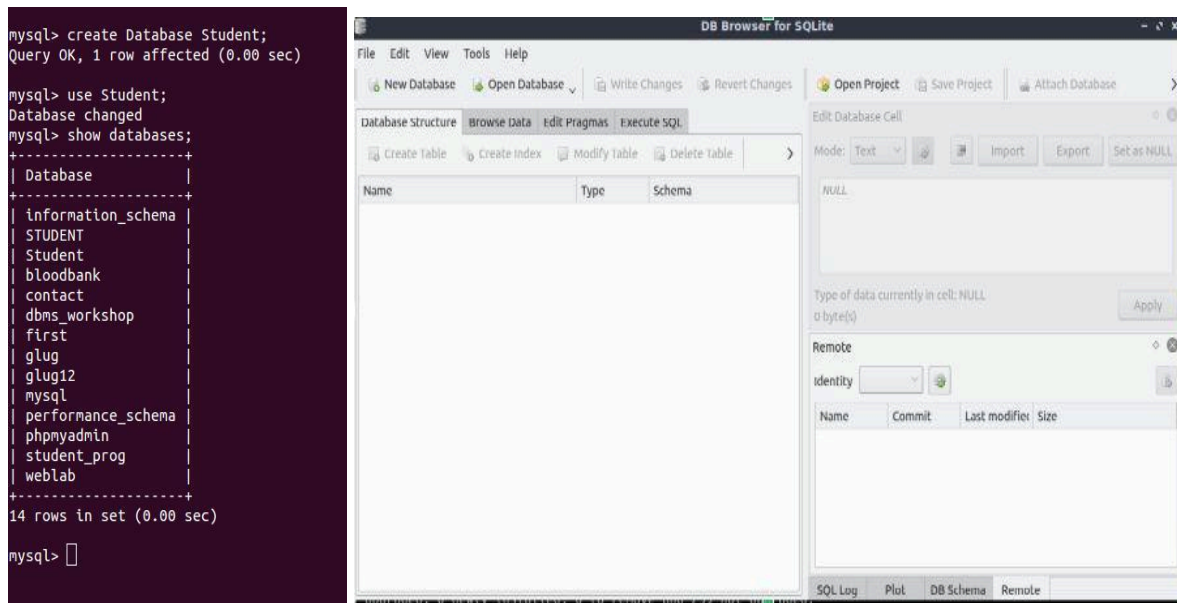
Specifications:

- Type: Computer Vision Library
- Language Support: Python, C++
- 5. Visual Studio Code (VS Code)



- **Used for coding, debugging, and project development.**
- **Provides an easy development environment.**

6. Database Management System

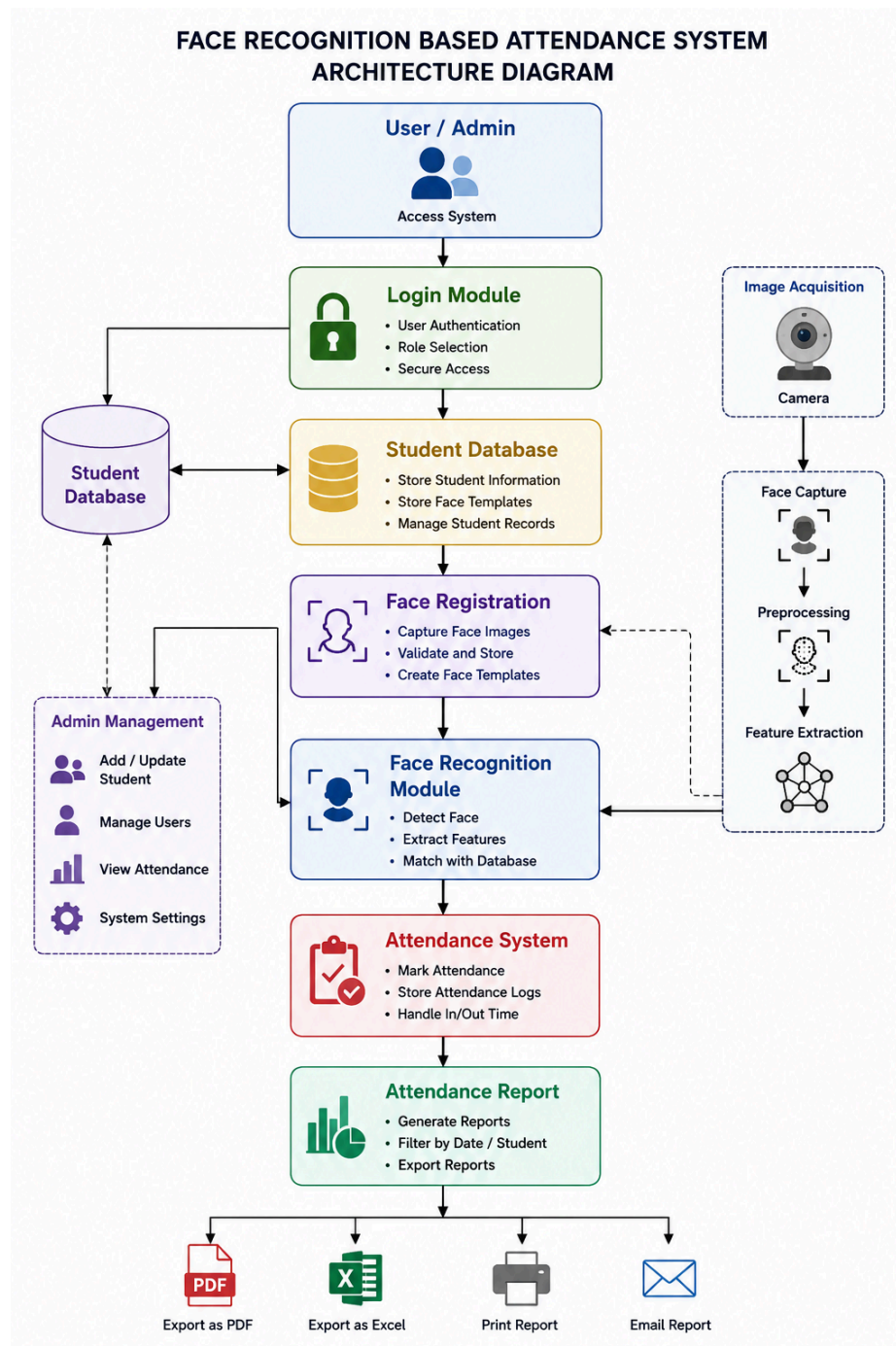


- **Stores student details and attendance records.**
- **Maintains digital attendance data securely.**

Specifications:

- **Database Type: SQLite / MySQL**
- **Features: Data Storage, Record Management**

3.3.2 System Diagram (Circuit connection diagram/Software architecture/product architecture with description)



Chapter 4

Implementation

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    "framer-motion": "^12.38.0",
    "idb": "^8.0.3",
    "lucide-react": "^1.7.0",
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"react-router-dom": "^7.13.2",  
  
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```

Chapter 5

Results and Analysis

User Testing & Feedback

Participants:

- 15 students and 2 faculty members participated in testing.

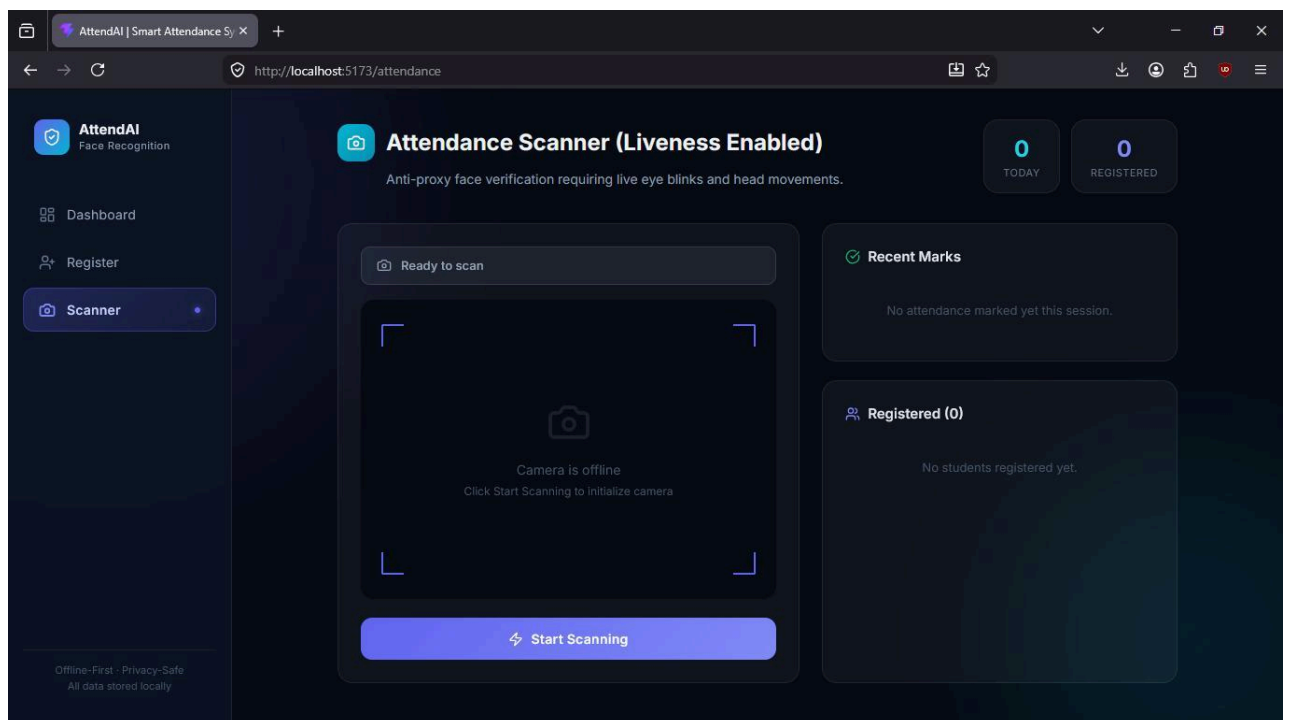
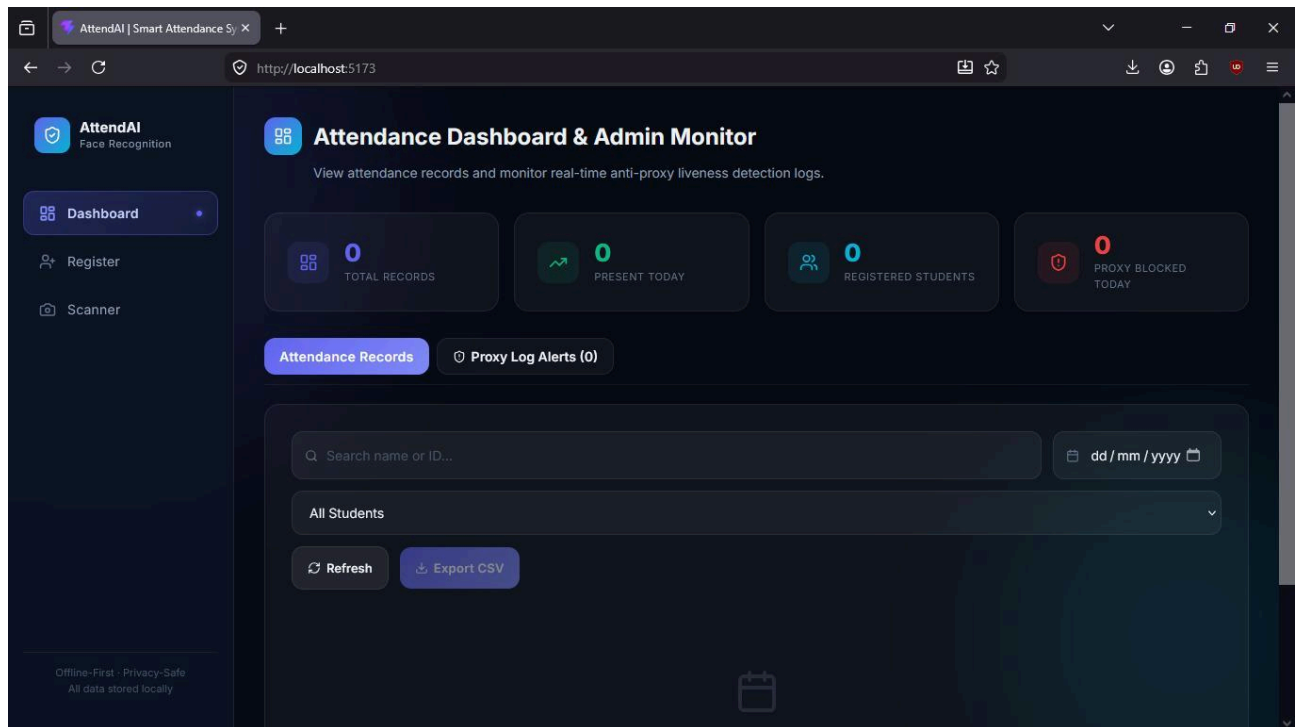
Quantitative Results:

- Attendance marking time reduced to approximately 2 seconds per student.
- Manual attendance time reduced from 10–15 minutes to less than 2 minutes.
- Face recognition accuracy achieved up to 98% during testing.
- Attendance records were updated instantly in the database.
- Proxy attendance cases were significantly reduced.

Qualitative Feedback:

- Students: “The system is quick and easy to use.”
- Students: “Attendance marking became much faster compared to manual methods.”
- Faculty: “Automatic attendance tracking reduced classroom workload.”
- Faculty: “The attendance reports are clear and easy to manage.”

- **Admin:** “Database records were stored accurately without mismatches.”
- **Users** suggested improving performance in low-light conditions and adding mobile support.



Chapter 6

The Smart Attendance System effectively addresses the limitations of traditional attendance methods by automating the attendance process using face recognition technology. The system improves accuracy, reduces manual effort, minimizes proxy attendance, and saves valuable classroom time. By integrating technologies such as Python, OpenCV, and database management systems, the project provides a secure, reliable, and user-friendly attendance solution. The developed software successfully demonstrated real-time attendance monitoring, digital record management, and efficient report generation. Overall, the project achieved its objectives of creating an accurate and automated attendance management system for educational institutions.

Future Work:

- AI-based advanced facial recognition for higher accuracy.
- Mobile application support for easier accessibility.
- Real-time notifications for students and faculty.
- Predictive attendance and absentee analytics.
- Cloud database integration for secure online storage.
- Automated parent notification system.
- Full ERP/LMS integration for institutional management.
- Improved performance in low-light conditions.

References

Research Papers / Books

- Michael J. Folk, Bill Zoellick, Greg Riccardi: *File Structures – An Object Oriented Approach with C++*, 3rd Edition, Pearson Education, 1998.
- K.R. Venugopal, K.G. Srinivas, P.M. Krishnaraj: *File Structures Using C++*, Tata McGraw-Hill, 2008.
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- Richard Szeliski: *Computer Vision: Algorithms and Applications*, Springer, 2011.
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Text Books / Study Materials

- <https://docs.python.org/3/>
- <https://opencv.org/>
- <https://realpython.com/face-recognition-with-python/>

Online Articles

- https://en.wikipedia.org/wiki/Face_recognition
- <https://towardsdatascience.com/face-recognition-how-lbph-works-90ec258c3d6b>
- <https://pyimagesearch.com/category/face-recognition/>

Annexures

Annexure A – User Feedback Forms

Feedback Form Sample

Sl. No User Type Feedback

1	Student	Attendance marking was fast and easy to use.
2	Student	The face recognition system reduced manual effort.
3	Student	Dashboard interface was simple and user-friendly.
4	Faculty	Attendance records were generated accurately.
5	Faculty	The system reduced classroom attendance time.
6	Faculty	Exporting attendance reports was very useful.

Annexure B – Iteration Notes

Iteration	Changes Made	Purpose
Iteration 1	Developed login and registration module	Basic system setup
Iteration 2	Added face detection and recognition	Automated attendance marking
Iteration 3	Improved database connectivity	Faster data storage and retrieval
Iteration 4	Enhanced dashboard design	Better user experience
Iteration 5	Fixed recognition errors in low lighting	Improved system accuracy
Iteration 6	Added attendance export feature	Easy report generation

Annexure C – Team Roles

Team Member	Role Assigned	Responsibility
Ganesh P	Project Leader & Backend Developer	Managed project planning, database integration, and overall system coordination
G Vinayak	Frontend Developer	Designed user interface, dashboard, and attendance display pages
Sumith S	Face Recognition Testing	& Implemented face detection module, testing, and debugging
Surabhi M	Documentation Presentation	& Prepared project report, documentation, PPT, and presentation materials